Table 15: 8-digit digit match accuracy with small dataset (1M samples) with RoPE, NoPE and Alibi.

	Int-add	Float-add	Fraction-multiplication	Scientific-add
RoPE	**0.97**	**0.99**	**0.33**	**0.99**
NoPE	0.78	0.98	0.29	0.96
Alibi	0.23	0.80	0.17	0.79

Table 17: Exact match of 0.1B models trained on integer addition, multiply and maximum respectively with various compositions of reverse formatting and index hints.

	Integer Addition				Integer Multiply				Integer Max			
	rev	rev + idx	no	idx	rev	reverse + idx	no	idx	reverse only	reverse + idx	no	idx
d9	**1.00**	0.93	0.98	0.41	**0.43**	0.00	0.13	0.00	**1.00**	0.99	**1.00**	0.99
d10	**0.80**	0.06	0.32	0.01	**0.13**	0.02	0.04	0.02	**1.00**	0.97	**1.00**	0.98

Table 18: Finetuning with PE, data format, and tokenizer modification will degrade the performance. The first two lines are a *naive finetuned* Llama and the original Llama *without finetuning*, which are the baseline. “1d” means using the *one-digit tokenizer* for numbers otherwise the *original tokenizer*. “rev” means *reverse* representation, where the integer parts are reversed. All the checkpoint we select by the lowest valid loss. The accuracy reported is the average “exact match” in each range. Metric “wld” is used to denote well-learned digit; “ppd” is used to denote performance-preserving digit.

	Integer Addition						Float Addition					
	S	M	L	XL	wld	ppd	S	M	L	XL	wld	ppd
FT	**0.95**	**0.65**	**0.12**	0.01	**4**	**12**	**0.96**	**0.71**	**0.27**	**0.08**	**5**	**17**
w/o FT	**0.95**	0.38	0.06	**0.02**	**4**	9	0.90	0.47	0.10	0.02	3	11
NoPE	0.67	0.04	0.00	0.00	3	5	0.37	0.06	0.00	0.00	0	0
NoPE + rev + 1d	0.89	0.35	0.06	**0.02**	3	9	0.81	0.38	0.09	0.01	0	11
NoPE + rev + pad + 1d	0.87	0.34	0.05	**0.02**	0	9	0.74	0.38	0.06	0.01	0	9
RoPE + 1d	0.93	0.59	0.05	0.00	4	9	0.33	0.30	0.06	0.01	0	9
RoPE + rev + 1d	0.40	0.20	0.04	0.00	0	7	0.35	0.30	0.09	0.02	0	11

	Fraction Multiplication (easy)						Scientific Notation Addition					
	S	M	L	XL	wld	ppd	S	M	L	XL	wld	ppd
FT	**0.43**	0.00	0.00	0.00	**1**	**3**	**0.09**	**0.08**	**0.02**	0.00	0	**4**
w/o FT	0.28	0.00	0.00	0.00	0	**3**	0.02	0.02	0.01	0.00	0	0
NoPE	0.14	0.00	0.00	0.00	0	**3**	0.00	0.00	0.00	0.00	0	0
NoPE +rev + 1d	0.25	0.00	0.00	0.00	0	**3**	0.01	0.02	0.01	0.00	0	0
NoPE + rev + pad + 1d	0.27	0.00	0.00	0.00	0	**3**	0.01	0.02	0.01	0.00	0	0
RoPE + 1d	0.09	0.00	0.00	0.00	0	1	0.08	0.03	**0.02**	0.00	0	3
RoPE + rev + 1d	0.06	0.00	0.00	0.00	0	1	0.08	0.02	0.01	0.00	0	**4**

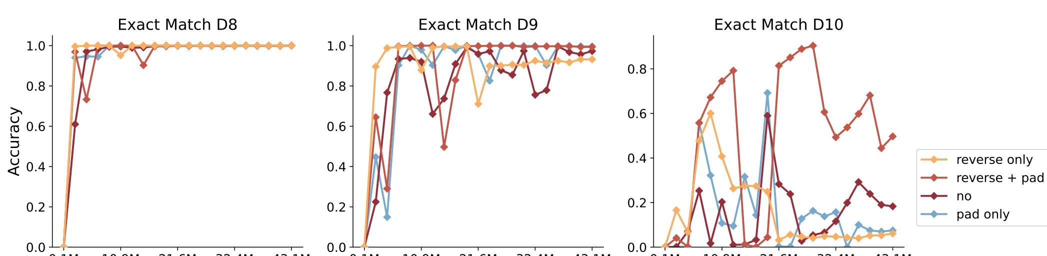

Figure 14: Exact match of 0.1B models trained on 1- to 8- digit integer addition with different compositions of reverse formatting and zero padding on 8- to 10- digit tests. X-axis is the number of seen training samples.

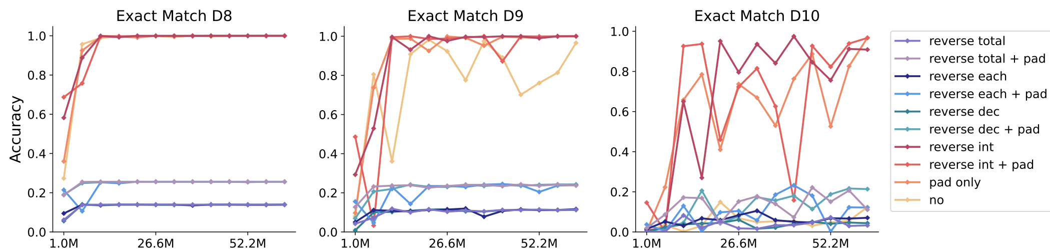

Figure 15: Exact match of 0.1B models trained on 1- to 8- digit float addition with different compositions of reverse formatting and zero padding on 8- to 10- digit tests. X-axis is the number of seen training samples.

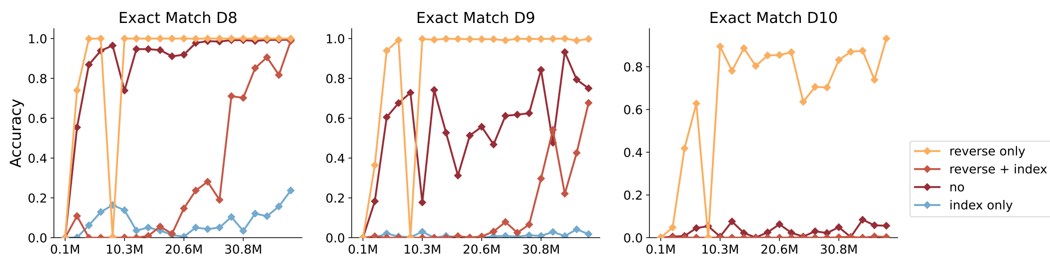

Figure 16: Exact match of 0.1B models trained on 1- to 8- digit integer addition with different compositions of reverse formatting and index hints on 8- to 10- digit tests. X-axis is the number of seen training samples.

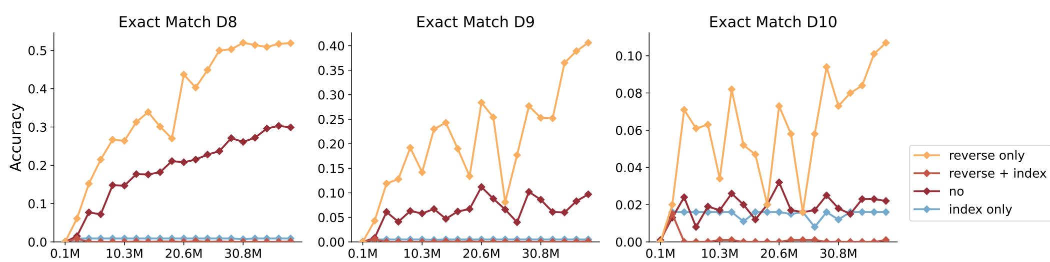

Figure 17: Exact match of 0.1B models trained on 1- to 8- digit integer multiplication with different compositions of reverse formatting and index hints on 8- to 10- digit tests. X-axis is the number of seen training samples.

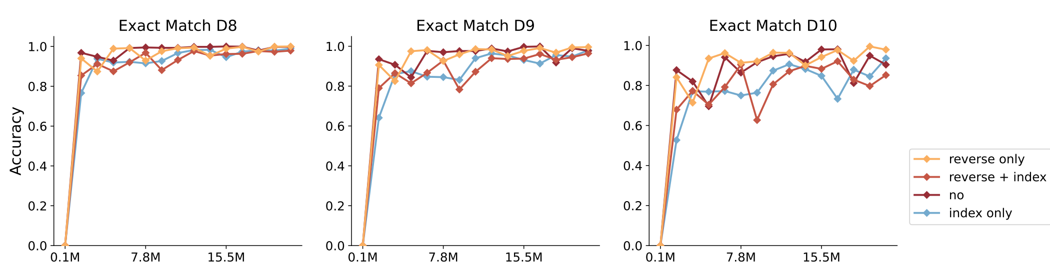

Figure 18: Exact match of 0.1B models trained on 1- to 8- digit integer maximum with different compositions of reverse formatting and index hints on 8- to 10- digit tests. X-axis is the number of seen training samples.

Table 19: Maximum length of each task that 2k context window can afford with RF-CoT

	Add	Sub	Multiply	Floordiv	Mod	Max	DigitMax	GetDigit	Length
Integer	20	20	12	20	6	100	17	100	34
Float	6	5	4	-	-	50	-	100	-
Fraction	3	2	3	-	-	20	-	-	-
Scientific	3	3	3	-	-	100	-	-	-